UNIVERSITY OF EDINBURGH
Business School

Experiments & Quasi-Experiments

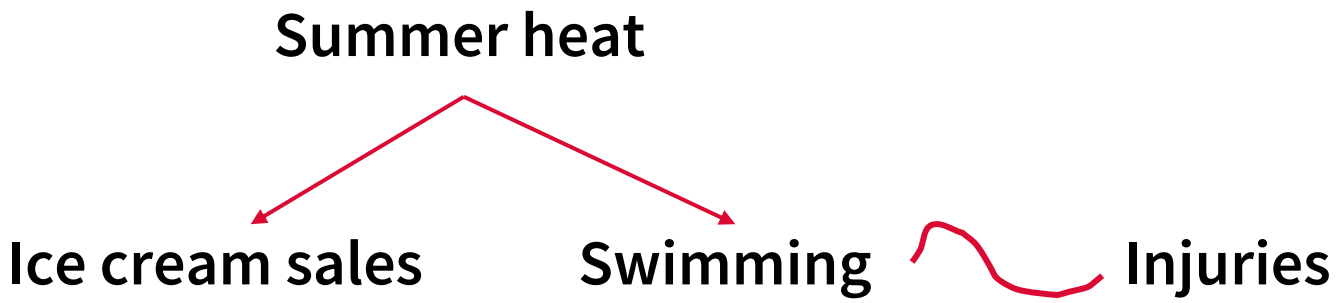


Han Zhang
Marketing Group, UEBS
hzhang21@ed.ac.uk

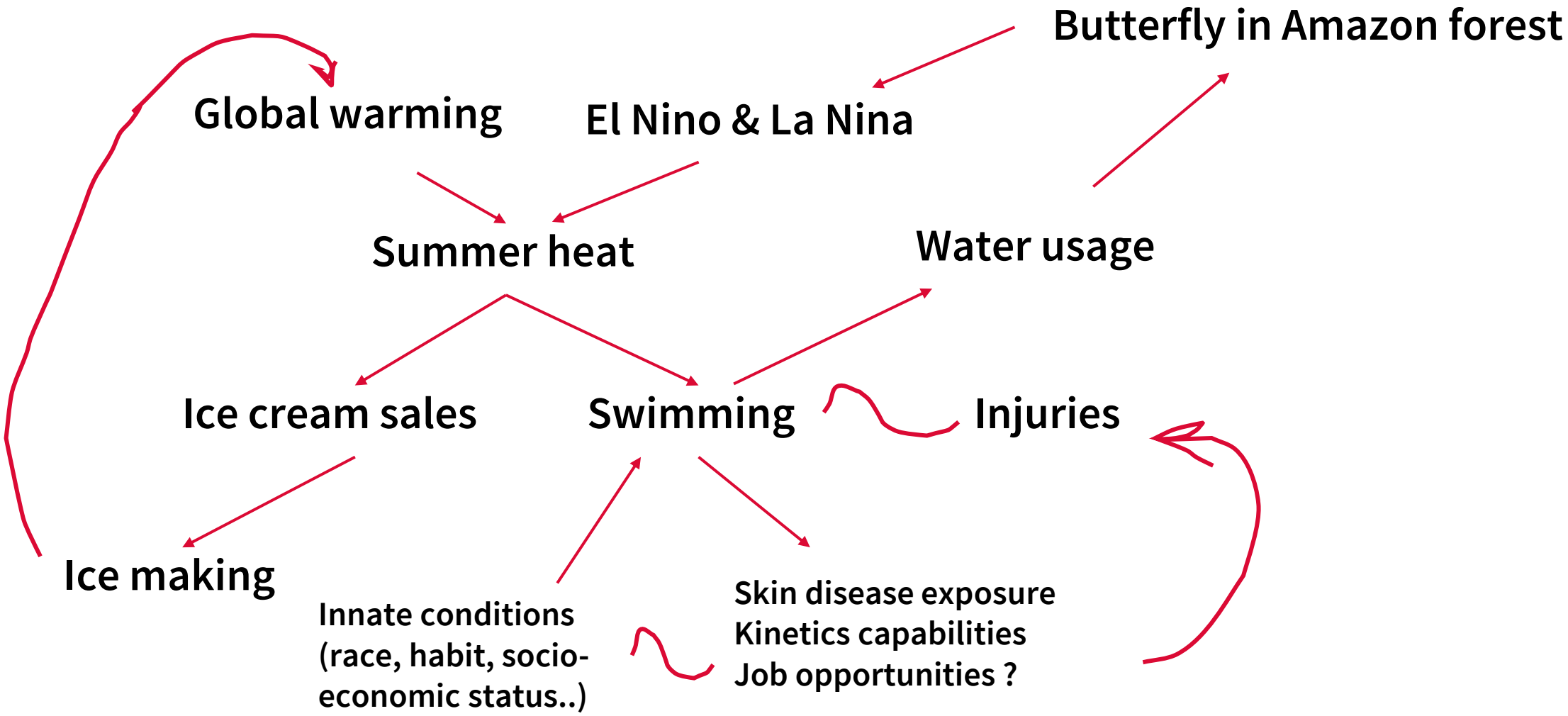
Causal Model

$$X \rightarrow Y$$

e.g. Ice cream sales → Swimming pool injuries (?)

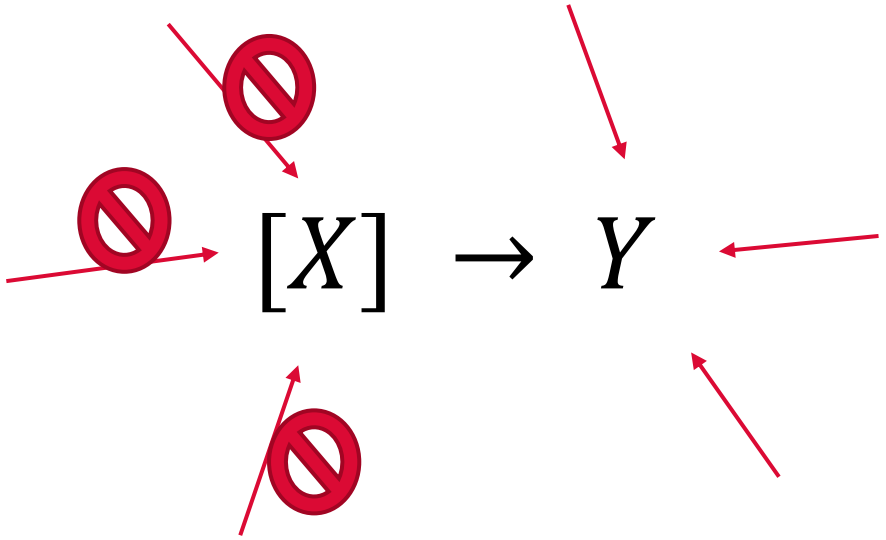


Causal Model



Experiments

Randomised Controlled Trial (RCT) is the gold standard for causal identification.



$$X_{=0}|D \rightarrow Y_0|D$$

$$X_{=1}|D \rightarrow Y_1|D$$

Effect of X changed from 0 to 1, conditional on D, is Y1 – Y0.

Experiments



(a) Control group (AC). (b) Treatment group (AT).

Fig. C.1. Images used to manipulate prior sponsorship level.

Experiments

The effect = mean(Y_AT) – mean(Y_AC)

The significance is tested using a ‘T-test’.

But actually, it is equivalent to the ‘b’ in this OLS regression: Y = bX + c
(for proof, ask any AI)

The basic idea: Effect = Mean[Group 1] – Mean[Group 0]

Stata example:

```
. sysuse auto
(1978 automobile data)

. tab foreign
```

Car origin	Freq.	Percent	Cum.
Domestic	52	70.27	70.27
Foreign	22	29.73	100.00
Total	74	100.00	

```
. ttest price, by(foreign)

Two-sample t test with equal variances
```

Group	Obs	Mean	Std. err.	Std. dev.	[95% conf.	interval]
Domestic	52	6072.423	429.4911	3097.104	5210.184	6934.662
Foreign	22	6384.682	558.9942	2621.915	5222.19	7547.174
Combined	74	6165.257	342.8719	2949.496	5481.914	6848.6
diff		-312.2587	754.4488		-1816.225	1191.708

```
    diff = mean(Domestic) - mean(Foreign)                         t =  -0.4139
H0: diff = 0                                     Degrees of freedom =       72

    Ha: diff < 0                 Ha: diff != 0                 Ha: diff > 0
 Pr(T < t) = 0.3401         Pr(|T| > |t|) = 0.6802          Pr(T > t) = 0.6599

. reg price foreign
```

Source	SS	df	MS	Number of obs	=	74
				F(1, 72)	=	0.17
Model	1507382.66	1	1507382.66	Prob > F	=	0.6802
Residual	633558013	72	8799416.85	R-squared	=	0.0024
				Adj R-squared	=	-0.0115
Total	635065396	73	8699525.97	Root MSE	=	2966.4

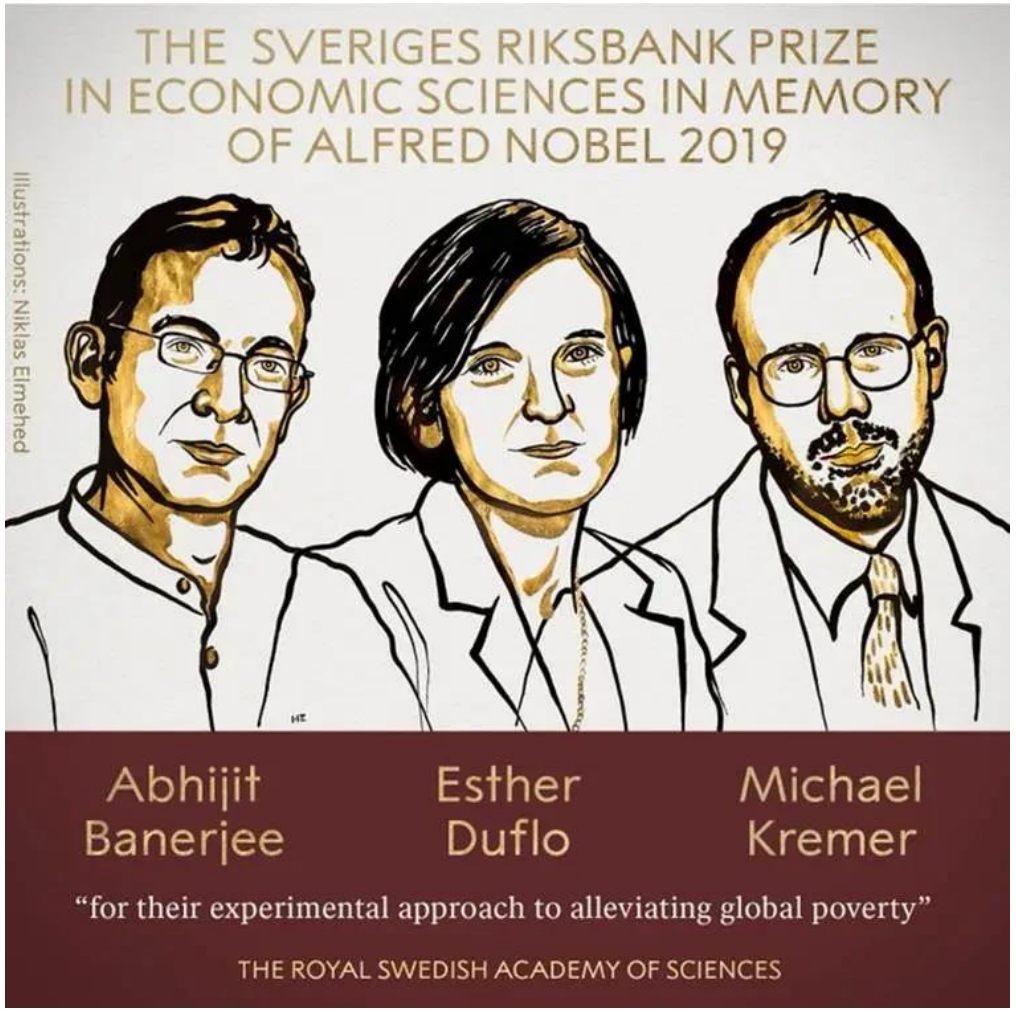
price	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
foreign	312.2587	754.4488	0.41	0.680	-1191.708	1816.225
_cons	6072.423	411.363	14.76	0.000	5252.386	6892.46

Quasi-Experiments

Unfortunately we cannot just randomly assign people to do things. Or can we?

Unless you are super rich or super funded (like those Nobel prize winners).

Actually, sometimes nature works in mysterious ways, and create a condition so similar to random assignment. We call them quasi-experiments.



Quasi-Experiments

Example of real random assignment in human history:

US military draft during WW2

Who serve was decided base on a national lottery, regardless of race, age, education, socio-economic status, skills, physiques, …

The economists are so joyed to identify the effect of military on people, as their enrollments were all randomly assigned.



https://www.nationalww2museum.org/students-teachers/student-resources/research-starters/draft-and-wwii

Quasi-Experiments

We cannot rely on random assignments, as rarely are there opportunities to dominate people. If it is close enough, we will take it.

Three common forms of such design:

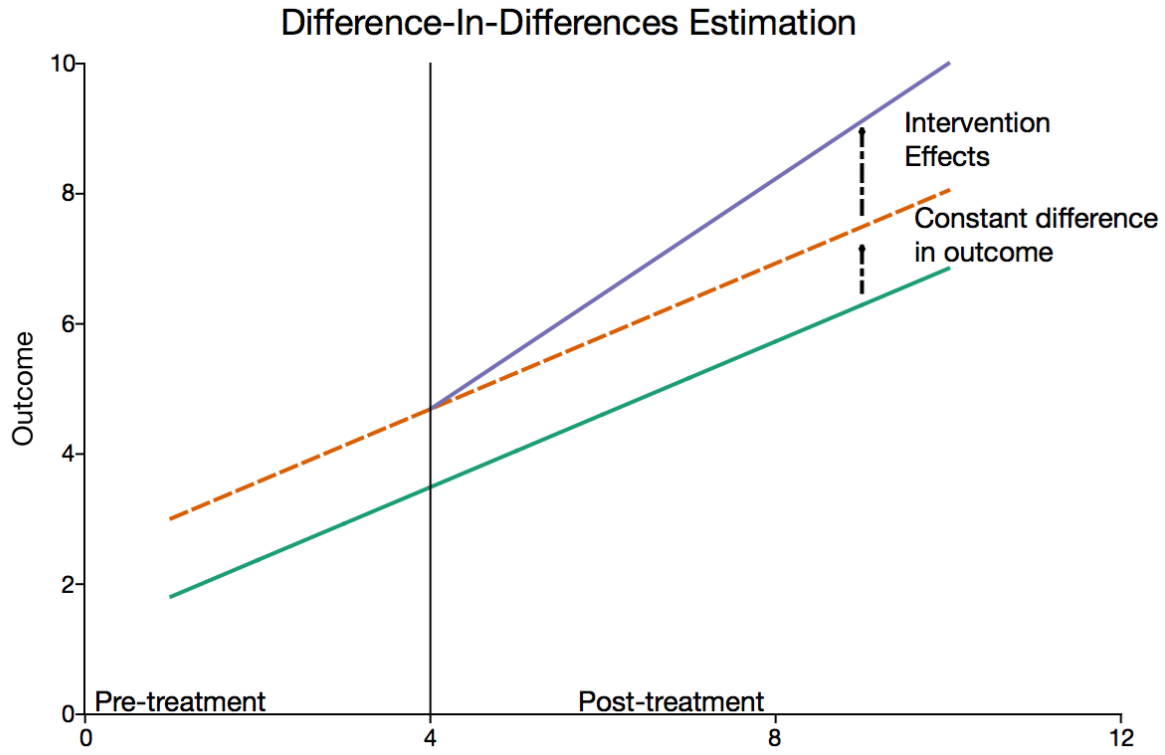
1. Difference-in-differences (DID)

2. Regression discontinuity design (RD, or RDD)

3. Instrument variable (IV)

1. Difference-in-differences (DID)

A policy will only affect [group 1] but not [group 0].
Had there been no such policy, [group 0] and [group 1] would've behaved similarly.



Effect* **= Effect[G1] – Effect[G0]**
= (Y[G1, post] – Y[G1, pre]) –
(Y[G0, post] – Y[G0, pre])

Policy is the only source of variation in this design.

* Again, this is equivalent to b in Y = b Treat*After + After + Treat + c, if you need proof ask AI.

2. Regression discontinuity design (RD, or RDD)

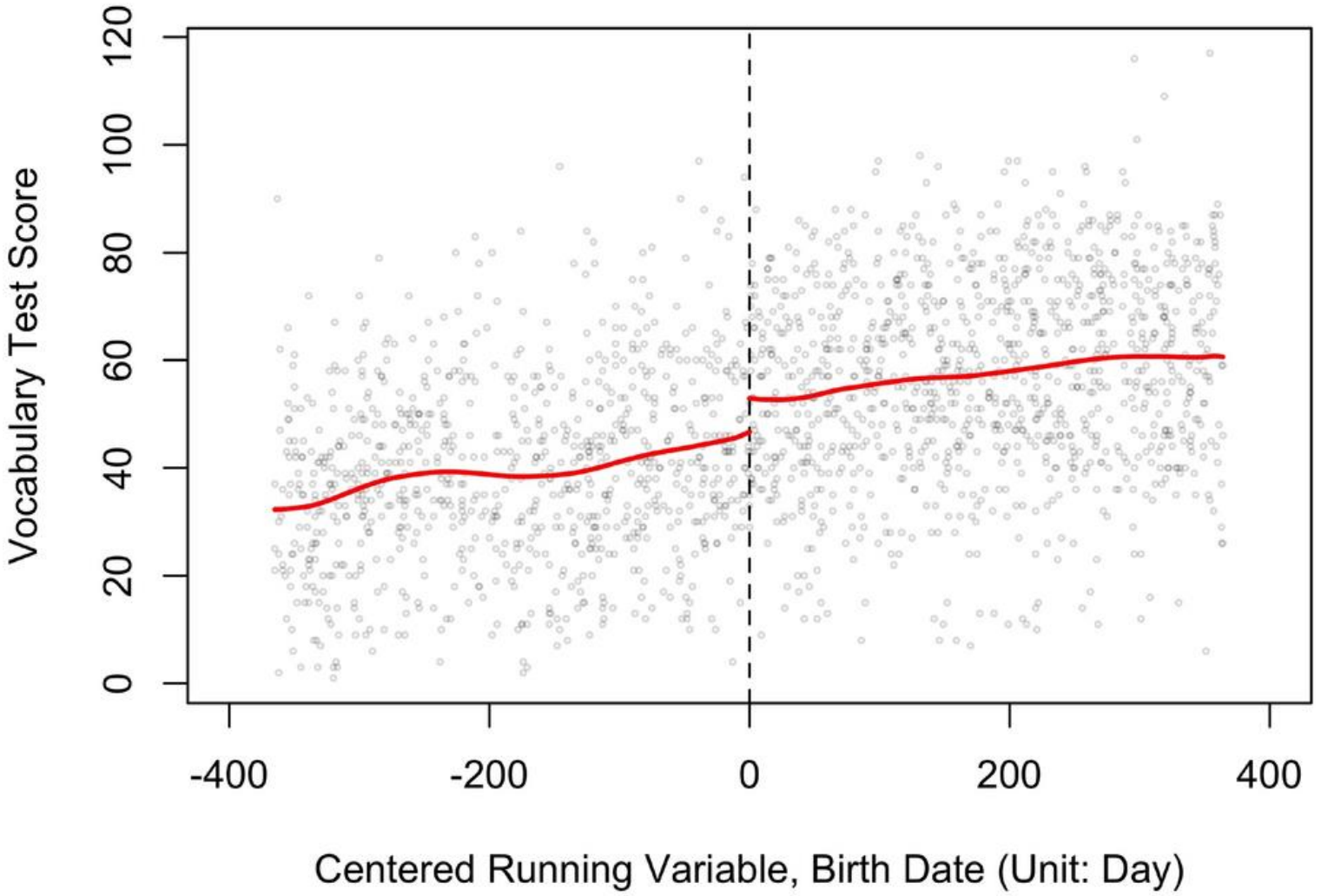
A numerical bar that is arbitrarily set.

Case 1: You must be over 18 to buy alcohol.
What's the difference between 17 yr 364 d old and 18 yr old?

Case 2: It is illegal to drive with over 0.1 g/100ml alcohol in blood.
What's the difference between 0.09g/ml and 0.10g/ml here?

Case 3: School admission based solely on test score.
Miss by 1 point and go to a different school.
How much difference in capabilities of students off by merely 1 point?

Example: Children born before 1st Oct can receive the education program on the same year. Delayed entry to the program jump-lift the vocabulary test score of children.

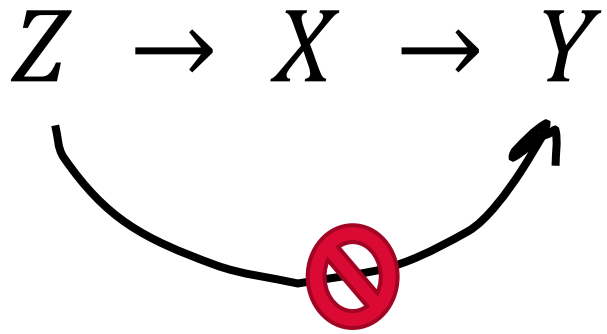


Effect* **= right – left (the jump)**

* Technically, you need to pick a band (a small value on X axis) to give the estimate of right and left endpoints. Which is why RDD is called a ‘design’, not a regression.

3. Instrument variable (IV)

A thing that leads to the change in X but not in Y.



If Z and Y change at the same time, it could only be via X.

X = smoke during pregnancy. Y = baby weight.

Problem: smoking mums probably have a hard life as well, which also hinders baby growth.

Z = cigarette tax.

$$\widehat{\beta_{OLS}} = \frac{\widehat{\mathrm{Cov}(X,Y)}}{\widehat{\mathrm{Var}(X)}} = \frac{\widehat{\mathrm{Cov}(Y,X)}}{\widehat{\mathrm{Cov}(X,X)}} \Rightarrow \widehat{\beta_{IV}} = \frac{\widehat{\mathrm{Cov}(Y,Z)}}{\widehat{\mathrm{Cov}(X,Z)}} = \frac{\bar{Y}_{Z=1} - \bar{Y}_{Z=0}}{\bar{X}_{Z=1} - \bar{X}_{Z=0}}$$

When assignment is not exogenous

Very often, being in [group 1] or [group 0] is decided by factors that also affects Y.

e.g. X = #Ad disclosure on social media. Y = like.

Problem: If you can get paid to do an #Ad, you are good at getting more likes already.

Luckily we also have three strategies to address the endogeneity in group assignment.

1. Matching (especially Propensity Score Matching, PSM)
2. Synthetic Control Method
3. Selection bias correction (Heckman)

1. Matching (especially Propensity Score Matching, PSM)

We care about effects on [group 1], who got treated (X=1).
Simply pick from [group 0] who are comparable to [group 1].
This is called Matching.

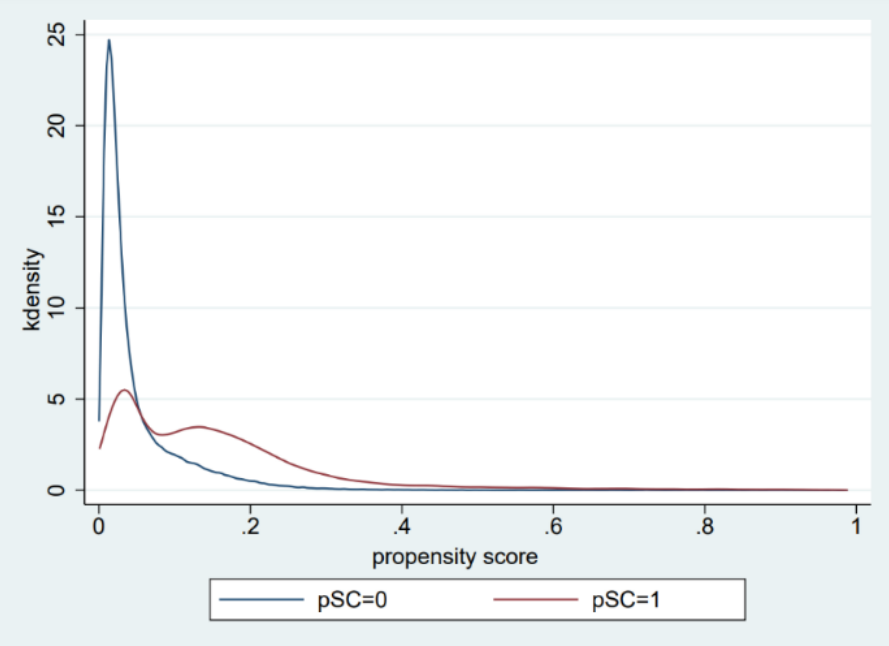
1:1 matching: For each sample in [group 1], pick the most similar one from [group 0], in terms of a given set of variables D.

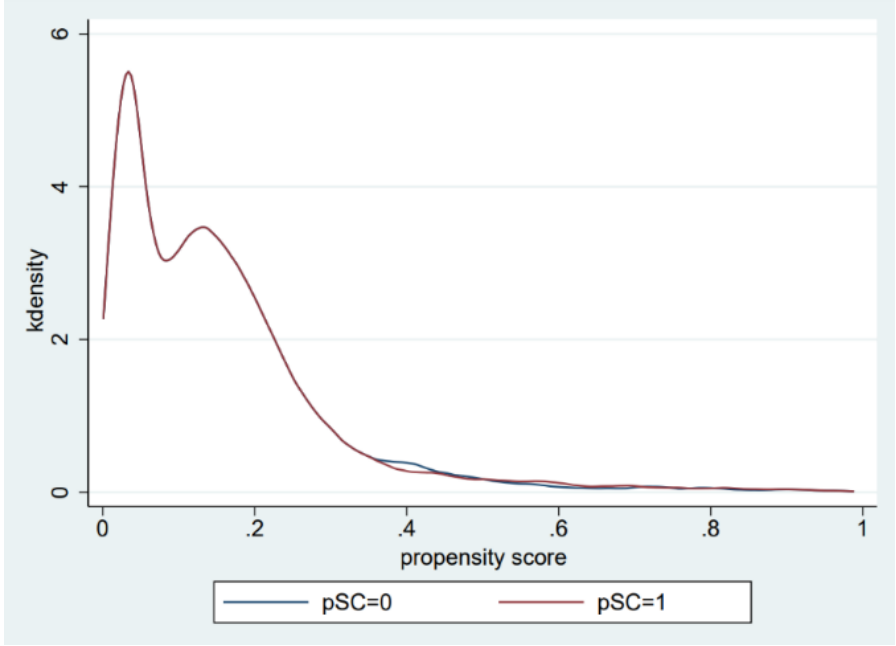
1:N matching: set a Calibre and accept all samples within the band.

Not good. Because it is actually hard to find a doppelganger.
Hence: Propensity Score Matching (PSM).

Propensity Score Matching (PSM)

Use Probit regression to estimate probability to be treated. $p(X|D) = \Phi(\gamma D)$
This gives the propensity score of a sample.
If [group 0] have the same propensity score distribution as [group 1], then we consider them comparable.





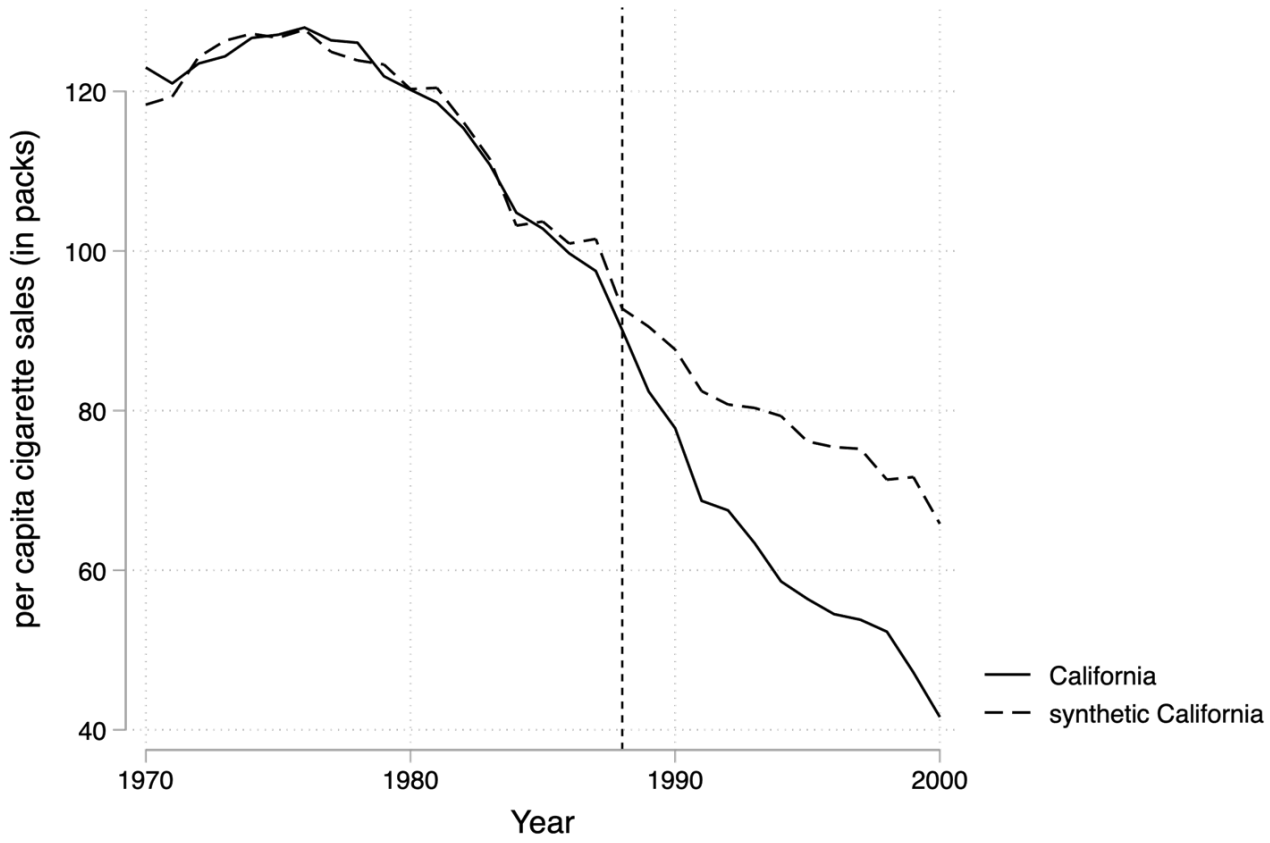
2. Synthetic Control Method

There is no [group 0] to compare to.

e.g. California was the only state to issue such policy.

Idea: Use a linear combination of other states to fit the outcome in pre-treatment years.

And hope that the trajectory can represent (counterfactual) post-treatment years as well.



3. Selection bias correction (Heckman)

Under certain assumptions, there is a correction term we can add to the regression that cancels out this selection bias.

The conditions are highly ideal and almost impossible to verify.
So JUST USE IT NOQA.

However, you will need at least 1 instrument variable to do this.

(Pro: if you use this, reviewers usually will only consider the eligibility of your IV and ignore other problems.)

Summary

1. Experiment (RCT) is gold standard in causal identification.
Experiment result is Mean[Treated] – Mean[Control].

3. Quasi-Experiments. Natural designs where only some are treated.
DID – point of execution; RD – numeric bar; IV – any exogenous source of variation.

The Effect is always simply a difference between two groups.

4. Strategies to address selection bias. PSM; Synthetic control; Correction term.

For more on methodology, please refer to: Goldfarb, A., Tucker, C., & Wang, Y. (2022). Conducting Research in Marketing with Quasi-Experiments. *Journal of Marketing*, *86*(3), 1-20. https://doi.org/10.1177/00222429221082977

Thank you

